

Challenging Behavior Profiles in Children with Co-Occurring IDD and PMS or FXS: A Multi-Method Approach to Inform Intervention Targets

Haley Zeri¹, Jason Cano¹, Edith Ocampo¹, Noah Torres¹, Niramay Patel¹, Madison Nava¹, Nathan Call², Joanna L. Mevers², Jennifer M. Hodnett², Lawrence Scahill², Elizabeth Berry-Kravis¹, Allison Wainer¹, Latha Valluripalli Soorya¹

¹Rush University Medical Center, Chicago, IL, ²Emory University, Atlanta, GA



Background

- **Challenging behaviors** are behaviors (e.g., Aggression, Self-Injury) that can impact those displaying them (i.e. daily functioning) and their caregivers.¹
- Here, we aim to understand challenging behaviors in children ages 2-12 years old with **Phelan-McDermid Syndrome (PMS)** or **Fragile X Syndrome (FXS)**.

We used the following measures to understand challenging behaviors:

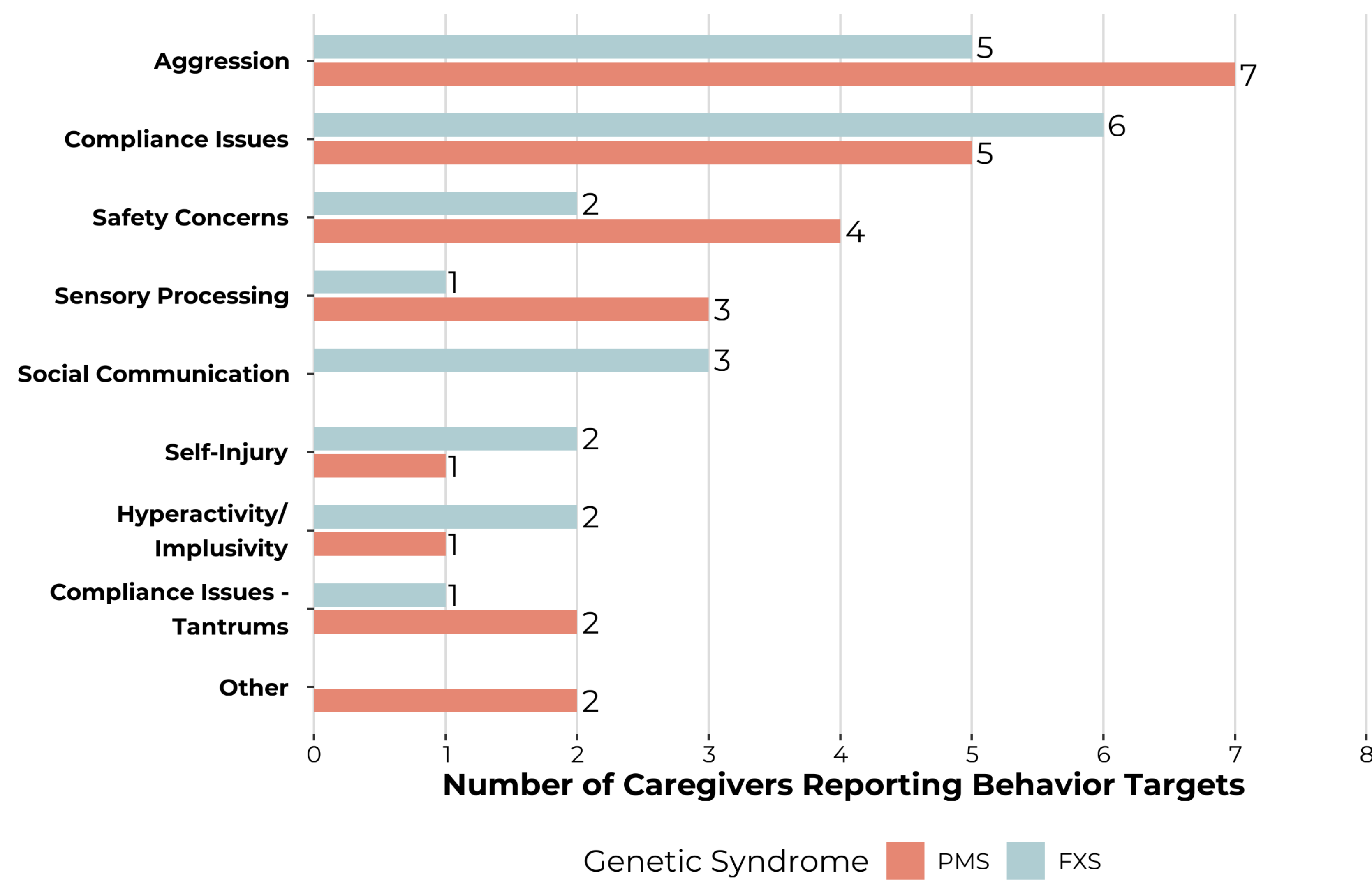
1. **Behavior Problems Inventory (BPI-01):** a behavior checklist of challenging behavior based on parent report.²
2. **Parent Target Problems (PTP):** an open-ended caregiver interview of high-priority concerns to inform treatment targets.³

Methods

- **23 caregivers** (Female: $n = 20$) of children with PMS ($n = 12$) or FXS ($n = 11$) and intellectual disability (IDD) ages 2-12 years old ($M = 7.65$, $SD = 3.19$) completed the BPI-01 and PTP as a part of a behavioral clinical trial (NCT06139172).
- The majority of children were male ($n = 16$) and had a Mild Intellectual Disability (ID) (Mild: $n = 11$, Moderate: $n = 8$, Severe: $n = 4$)
- **48 high-priority concerns** from all PMS and FXS caregivers were grouped into **9 behavior categories**⁴ (Figure 1).
- We calculated the percentage of caregivers who reported **at least one behavior** related to Aggression, Stereotypies, Self-Injury on both measures (PTP and BPI-01)⁵ (Figure 3).

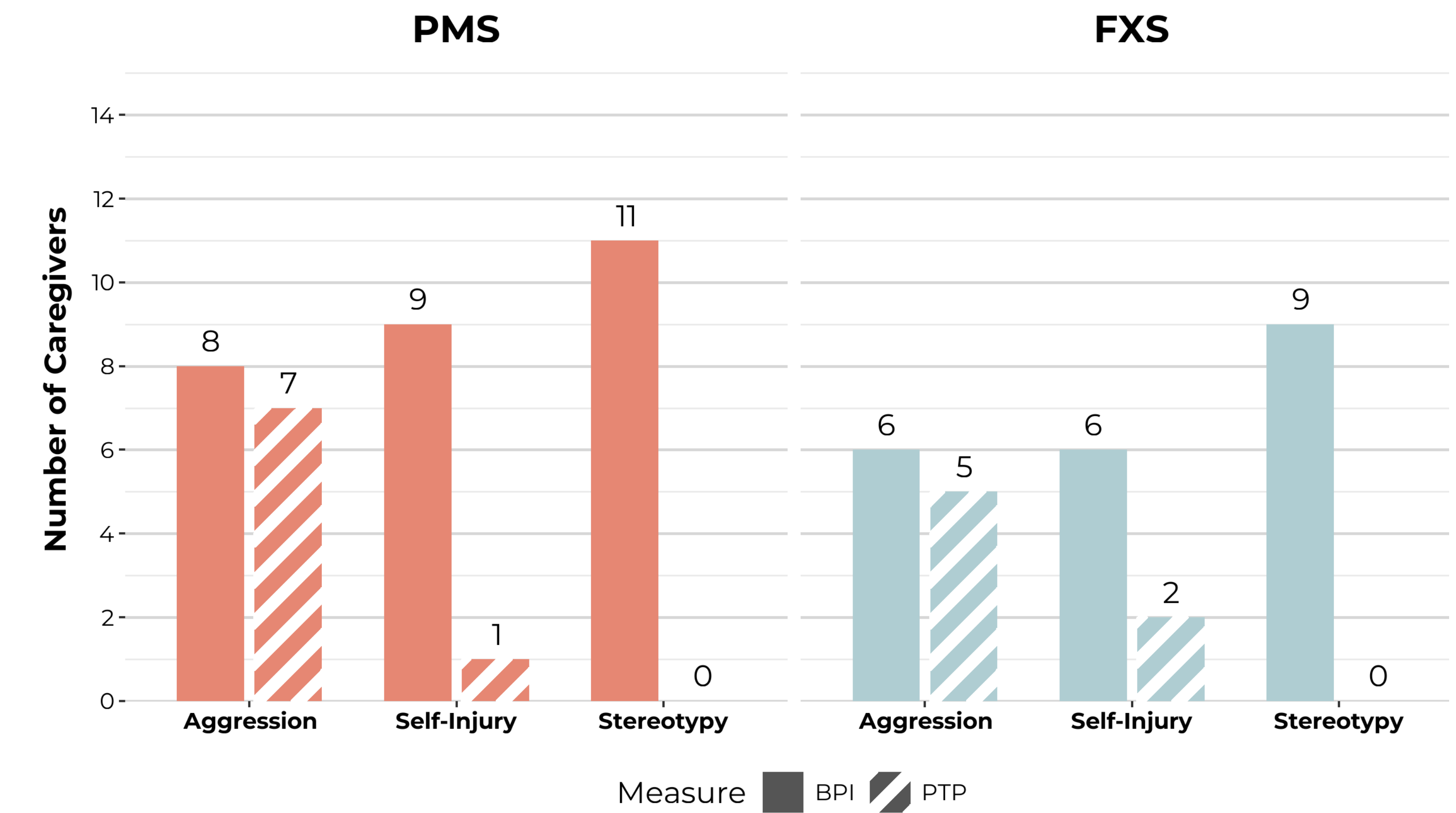
Results

Figure 1. PTP: Caregivers' High-Priority Concerns to Inform Intervention Targets



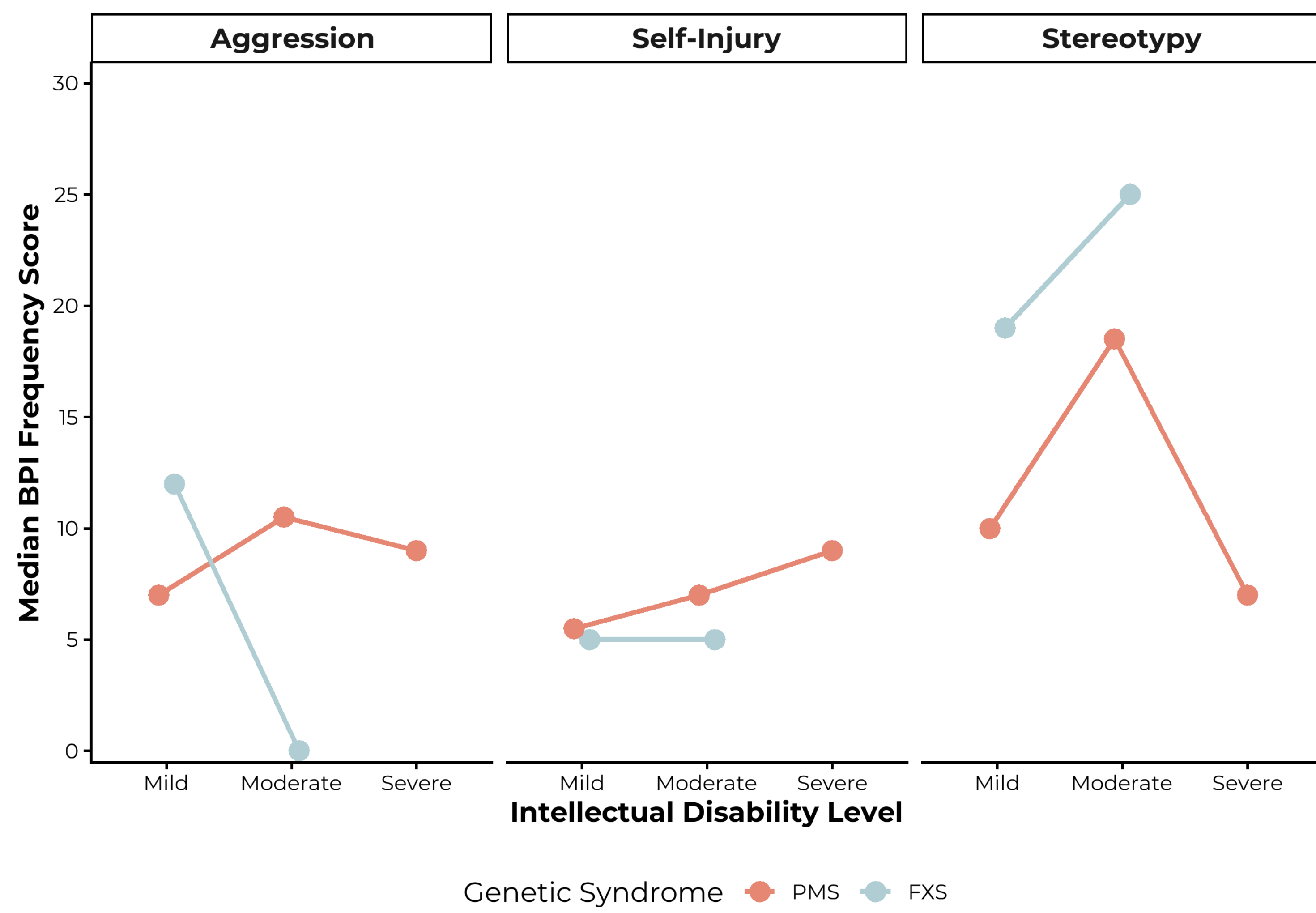
Aggression and **Compliance Issues** were among the top **high-priority concerns** caregivers identified for PMS and FXS children.

Figure 3. Caregiver-Reported Concerns across Measures: PTP (High-Priority Concerns) and BPI-01 (Behavior Frequency/Severity Checklist)



Self-Injury and **Stereotypy** were **less likely** to be selected as **high-priority concerns** (PTP) by caregivers across **both conditions**.

Figure 2. BPI-01: Median BPI Frequency Score by Subscale and Intellectual Disability Level



Median BPI-01 **Self-Injury** subscale scores appear to **increase** with ID level in PMS children.

Conclusion

- **Self-Injury** and **Stereotypy** were **less likely** to be selected as **high-priority concerns**, **although are reported to be frequent** by both PMS and FXS caregivers.
- Multi-method approaches that **differentiate characterization** (e.g., type, severity, frequency) **from caregiver-selected treatment targets** are crucial.
- The relationship between **ID level** and **frequency** of challenging behaviors suggests **different** patterns across **PMS** and **FXS**.
- **ID level** appears to be associated with **increasing frequency** of **Self-Injury** in **PMS** but not **FXS**, though additional research and replication in larger samples is needed.



WINGS Website

We would like to thank all of the families who made this research possible, the team at Emory University for their collaboration and expertise, and the NIH National Institute of Child Health and Human Development (NICHD) for funding this study (1R01HD107522).

ClinicalTrials.gov NCT06139172



References